Implementation of a VR- and LLM-Driven Virtual Patient Interaction System

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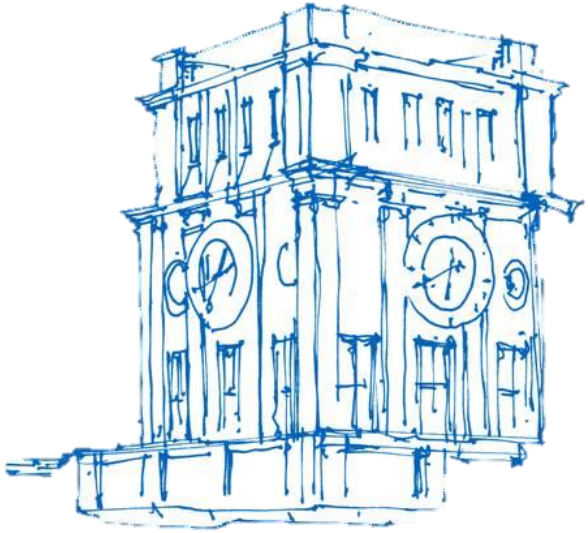

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Chair for Human-Centered Technologies for Learning

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Motivation

- Communication is central to medical practice
- Traditional training is costly and hard to scale
- Difficult conversations require repeated practice
- Immersive and adaptive learning environment

The Core —— Personalized Learning

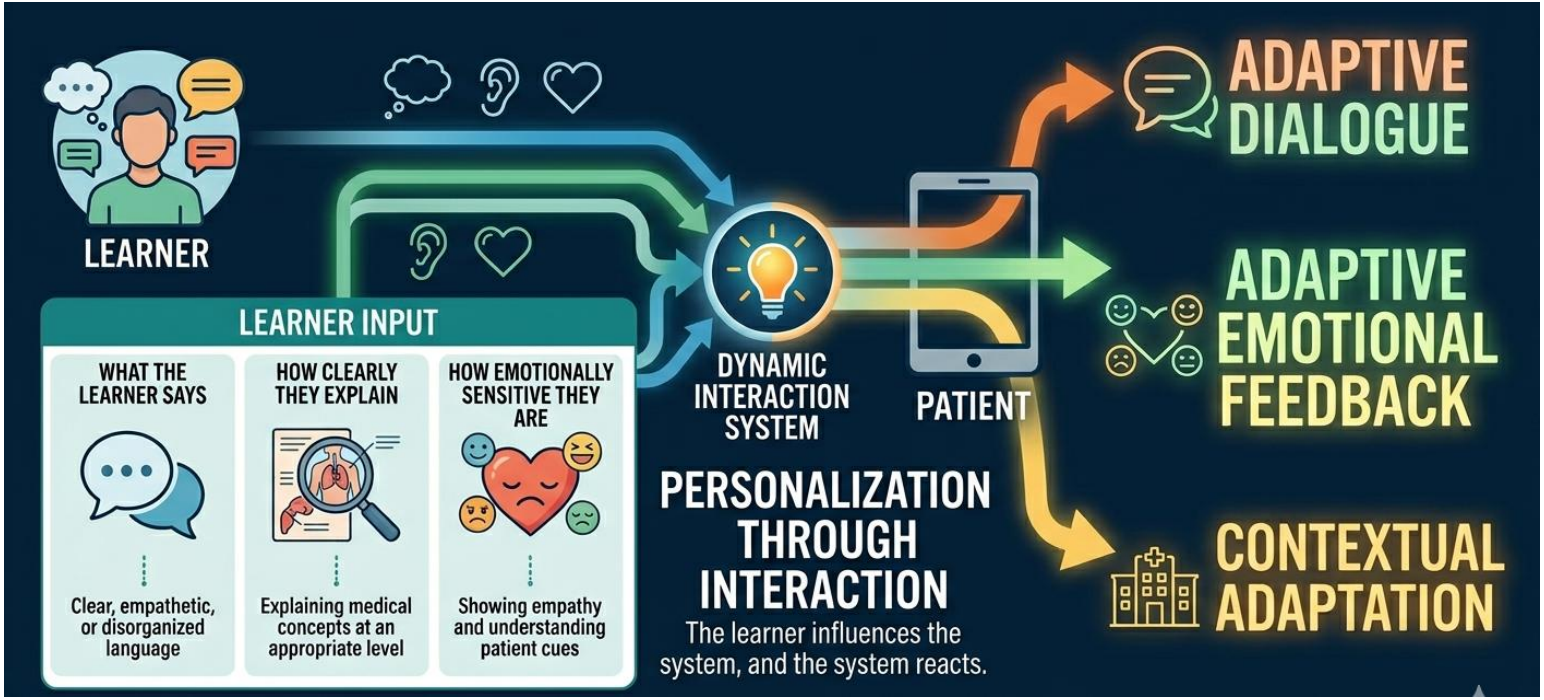


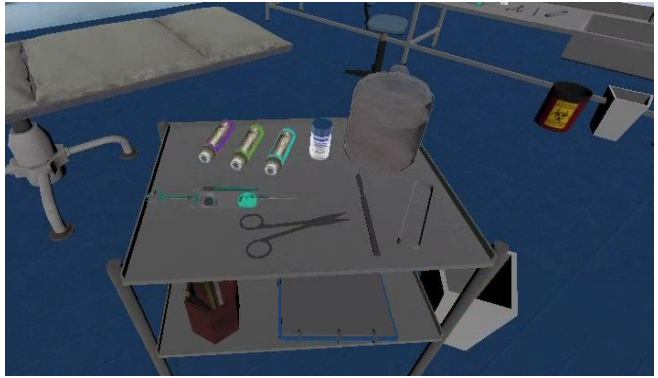
Figure: https://www.collidu.com/presentation-personalized-learning

VR for Medical Communication

SPIKES as the Training Framework

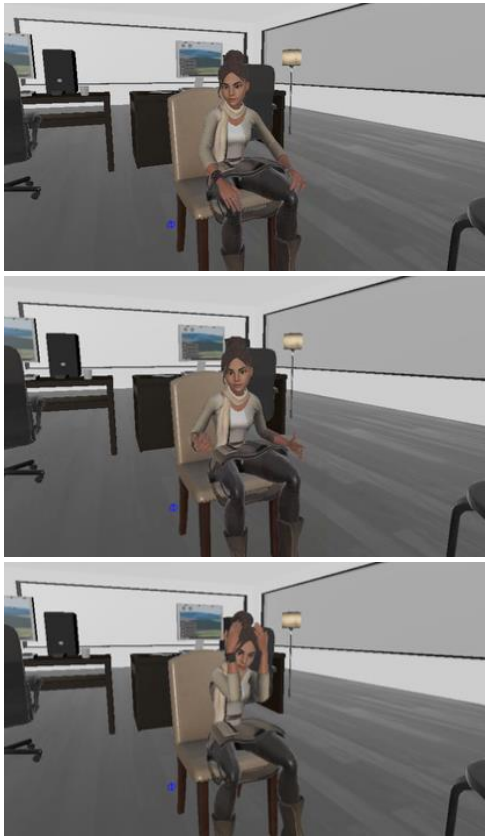
Learning Scenarios in Virtual Reality

Learning Scenarios in Virtual Reality



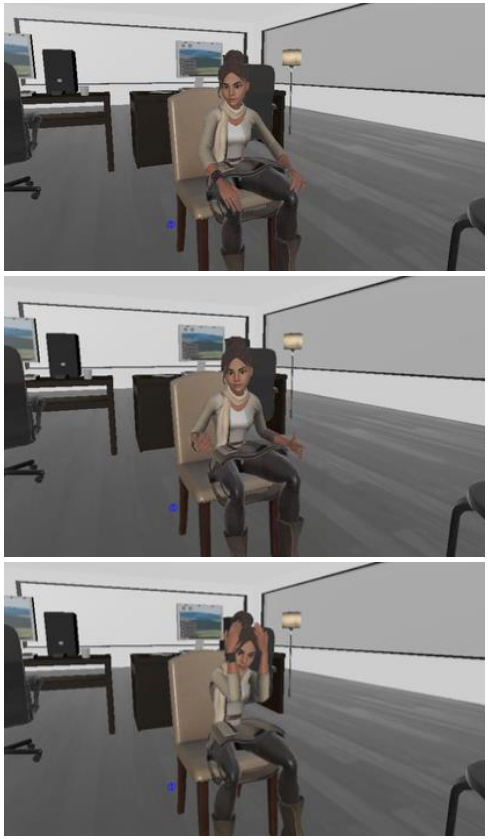
Emotional Modeling of the Virtual Patient

```
// Simple example: Set different animation states based on
anxiety levels

        switch (anxietyLevel.ToLower())
        {
            case "mild":
                avatarAnimator.SetFloat("AnxietyLevel", 0.3f);
                break;
            case "moderate":
                avatarAnimator.SetFloat("AnxietyLevel", 0.6f);
                break;
            case "severe":
                avatarAnimator.SetFloat("AnxietyLevel", 1.0f);
                break;
            default:
                avatarAnimator.SetFloat("AnxietyLevel", 0.5f);
                break;
        }
```



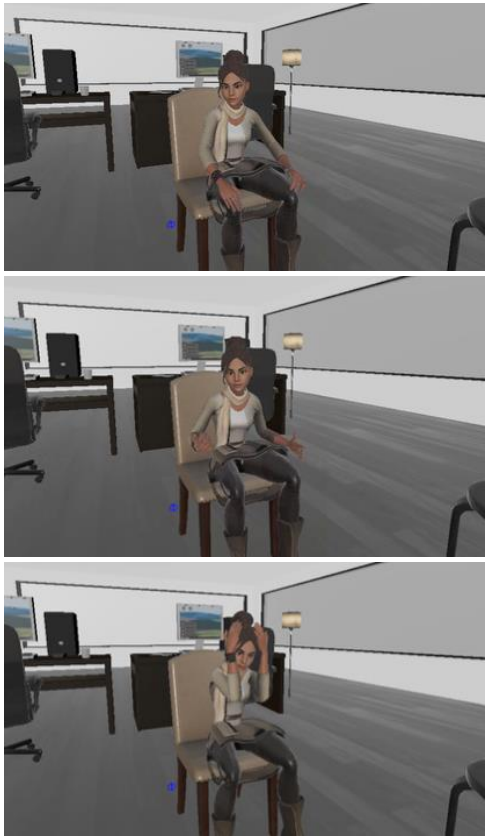
Emotional Modeling of the Virtual Patient

```
[Header("Positive vocabulary (reduces anxiety)")]
public List<string> positiveWords = new List<string>()
{
    "good", "great", "easy", "relaxed", "yes", ……
};

[Header("Negative vocabulary (increases anxiety)")]
public List<string> negativeWords = new List<string>()
{
    "bad", "hard", "serious", "no", ……
};

[Header("Reinforcing words (enhancing emotional impact)")]
public List<string> intensifiers = new List<string>()
{
    "very", "extremely", "absolutely", "really", ……
};

[Header("Weakening words (reducing emotional impact)")]
public List<string> softeners = new List<string>()
{
    "a little", "slightly", "somewhat", "a bit", ……
};
```



Emotional Modeling of the Virtual Patient

```
// Principle: The more anxious, the harder to rise further, but
easier falls.
        float attenuationFactor;
        if (deltaFromLLM > 0)
        {
            // When anxiety increases: the more anxious, the
harder to rise
            attenuationFactor = 1 - current_anxiety;
        }
        else
        {
            // When anxiety decreases: the more anxious, the
easier to fall
            attenuationFactor = current_anxiety;
        }
        float adjustedDelta = deltaFromLLM * attenuationFactor;
```



Technical Components

GPT-4o

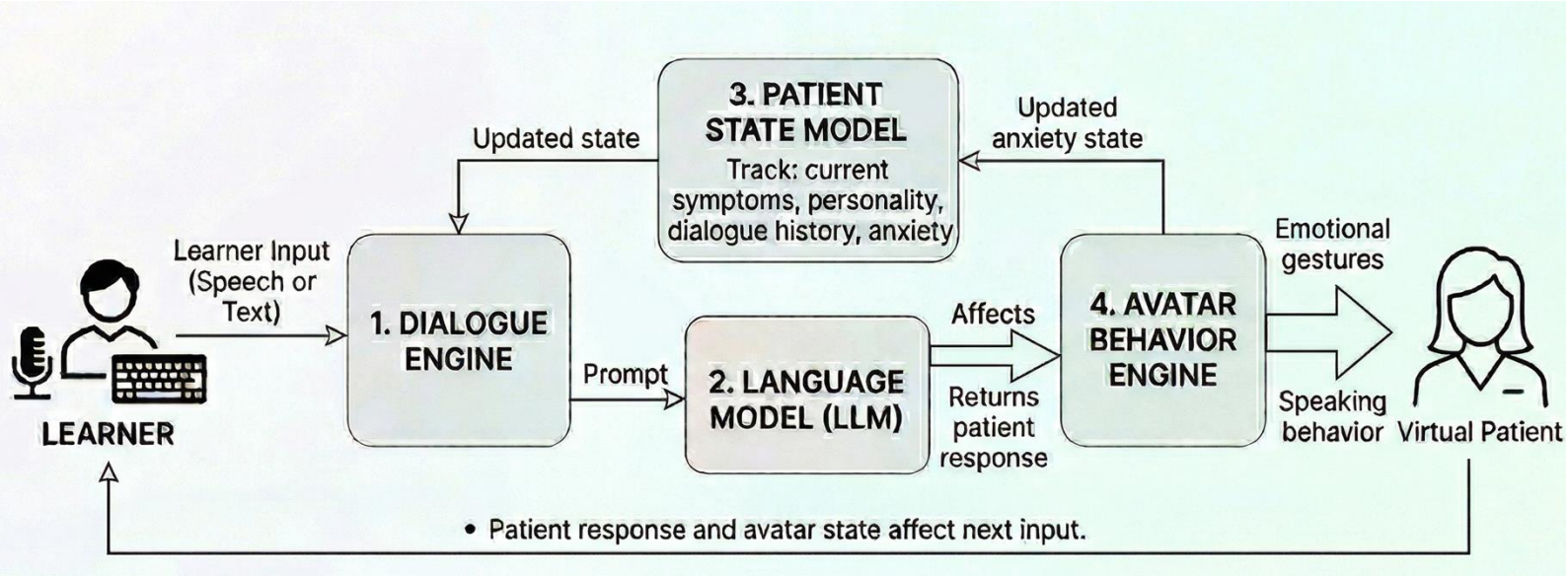
```
curl https://api.openai.com/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $OPENAI_API_KEY" \
  -d '{
    "model": "gpt-4o",
    "messages": [
      {
        "role": "system",
        "content": "You are a helpful assistant."
      },
```

Technical Components

- ChatGPT, GPT-4o + OpenAI API
  - More tokens and lower latency

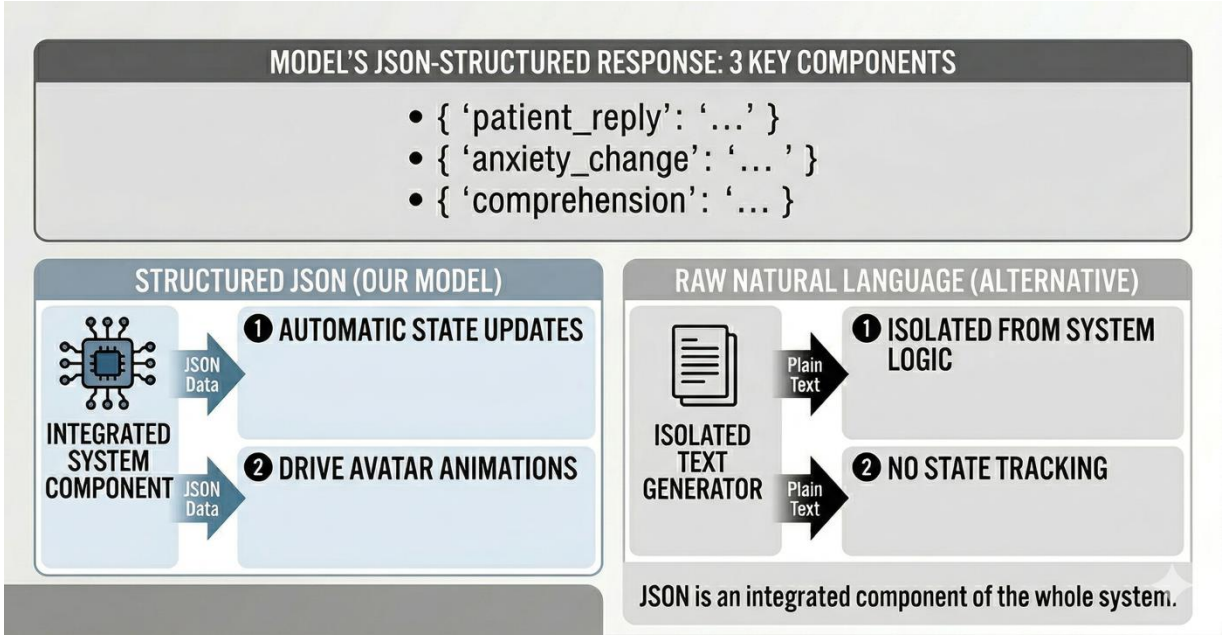
- Ready Player Me's avatar + Oculus lipsync + Mixamo animation
  - Concurrent speech speed
  - Real-time input and output of subtitles
  - Rich collection of ready-to-use character animations

System Architecture



The virtual patient is not stateless. What the user/learner says affects the virtual patient state, and the state affects the next response.

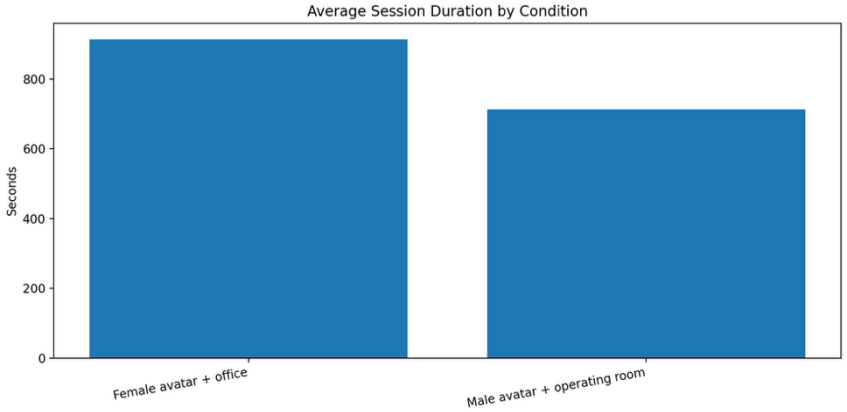
Structured In- and Output

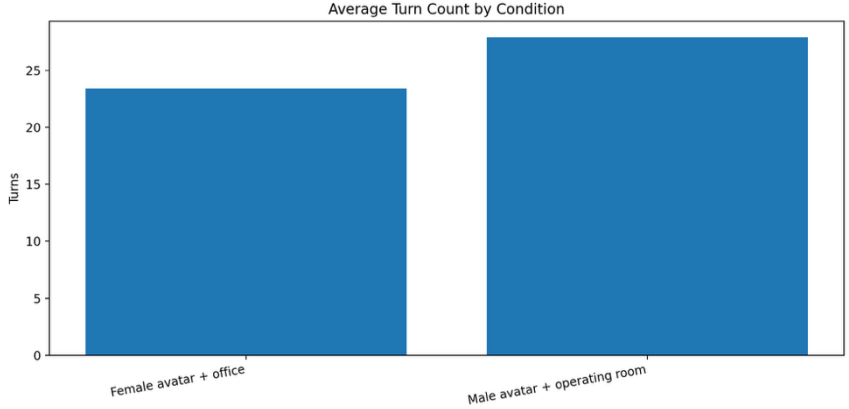


Structured in- and output enable reliable state update and avatar control.

Implementation of Personalized Learning

Metric	Female avatar + office	Male avatar + operating room
Average session duration	914.07 s	713.20 s
Average user word count	595.0	543.4
Average response time	24.70 s	18.65 s
Average turn count	23.40	27.93
Average sentiment score	0.32	0.36





Conclusion and Future Work

- Personalized Learning in Medical Education
  - SPIKES as a hidden design framework
  - Virtual patient system that combines VR and LLM
- LLM-driven Adaptive Interaction
  - Human-AI interaction depends on user's behaviors
  - Dynamic prompts and state update
- Future Work
  - Develop automated SPIKES-based assessment
  - Richer patient cases
  - Comprehensive emotional modeling

Thank you for your attention!

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